# AMLD Security 4

## Federated Verified Memory Across Eight Application Domains

A DATA-MIND 2.10 successor to

*AMLD Security 3*

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Companion research manuscript v1.0

September 2, 2026

The foundational treatise develops separation, containment, verification, and the Ocean gate. *AMLD Security 3* carries those ideas into eight application domains. This successor asks what changes when DATA-MIND 2.10 replaces a single shared-memory topology with a verifier-gated federation: one logically shared core, department-local BANK nodes, explicitly coupled neighbors, and local, coupled, core, or broadcast propagation. The verifier, FUTUREBANK, Sentinel, and Quarantine remain separate. The new mathematics concerns topology-safe truth, selective reuse, dependency closure, compartment-aware visibility, and experiments that can distinguish federation from mere duplication. No chapter claims that federation has already improved theorem-settlement performance or solved a deployed security problem.

# Companion Statement

This manuscript is complementary in a precise sense. The foundational older sibling, *Security by Separation, Verification, and Settlement* [5], moves from elementary security concepts toward AMLD. *AMLD Security 3* [6] then moves outward from AMLD toward eight application domains. The present successor preserves those domains and asks how their formal obligations change under DATA-MIND 2.10's federated verified-memory topology [9].

| | Foundational treatise | Security 3 | Security 4 |
|---|---|---|---|
| Primary question | What defensive architecture should exist? | What could verified settlement do in eight domains? | What changes when trusted memory is federated? |
| Direction | Security $\to$ AMLD | AMLD $\to$ applications | Federation $\to$ selective application memory |
| Main result type | Defensive theorem | Conditional application sketch | Topology invariant, visibility obligation, or experiment |

The shared writing rule remains *motivation first, formalism immediately after*. The shared security policy remains defensive: the mathematics is organized around auditing, containment, certification, repair, and controlled deployment, not around operational exploitation.

**Implementation status.** DATA-MIND 2.10 is represented by an accepted-for-integration-testing architecture decision, a public architecture marker, executable federation primitives, P/R/I/C coordinator wiring, unit tests, and a successful federation CI run [10, 11]. Named endpoints for Quotient Hunter, Professor, Proof Compass, Child, Presentation Manager, Learner, Sentinel, and Verifier are present in the topology; their full reasoning implementations are not thereby complete. Accordingly, this manuscript distinguishes implemented memory semantics from proposed application behavior.

# Abstract

An automated target prover, theorem prover, or metalogical decider becomes security-relevant when its target is a vulnerability, an invariant, a compressed state representation, or a certificate about a protected system. The central problem is not merely computational power. It is how to separate creative search from the authority to promote claims into trusted state. DATA-MIND 2.10 adds a second question: which verified departments should physically retain or logically see each accepted item? This manuscript gives a common verifier-gated federated formalism and applies it to eight domains: cryptography, vulnerability research, critical infrastructure, military and intelligence compartmentation, software correctness, scientific discovery, engineering optimization, and AI safety. Across the applications, four motifs recur: certificate-level settlement by the $P/R/I/C$ roles; separation of verified BANK, speculative FUTUREBANK, and security-quarantined state; topology-independent verifier soundness; future-sufficient quotienting; and Ocean-style regression gates for candidate machinery. We prove that verifier-gated federation preserves truth status independently of item location and state a dependency-visibility condition needed for sound reuse. We then derive domain-specific coupling patterns and a frozen-cohort experiment comparing core, broadcast, local, and coupled propagation. The results remain conditional: integration tests establish memory semantics, not a performance advantage, a practical cryptanalytic attack, a universal vulnerability finder, or global AI safety.

# Contents

# Part I

# The Shared Formal Spine

# Chapter 1

# From Intelligent Target Seeking to Verified Settlement

## 1.1 Targets, certificates, and environments

Let a target instance be

$$\mathfrak{I} = (c, \Gamma, \mathcal{E}, V, \mathcal{B}),$$

where $c$ is a claim or goal, $\Gamma$ is the admitted theory or specification, $\mathcal{E}$ is the declared environment, $V$ is an independent verifier, and $\mathcal{B}$ is a resource budget. Search may be learned, reflective, randomized, federated, or otherwise adaptive. Validity is narrower:

$$\mathrm{Accept}(\pi, c \mid \Gamma, \mathcal{E}) \iff V(\pi, c, \Gamma, \mathcal{E}) = 1.$$

Changing a search policy may change which certificate is found and how quickly it is found; it must not silently change the acceptance relation.

## 1.2 Five operational outcomes

For a formal target $c$, an AMLD-style process distinguishes

$$\mathsf{Settle}(c) \in \{P, R, I, C, \mathsf{UNKNOWN}\}.$$

Here $P$ supplies a verified certificate for $c$, $R$ supplies one for $\neg c$, $I$ supplies an independence certificate in a declared metatheory, and $C$ supplies an inconsistency certificate for the admitted assumptions. The fifth outcome records resource-bounded nonsettlement. It is crucial that

$$\mathsf{UNKNOWN} \not\Rightarrow c, \qquad \mathsf{UNKNOWN} \not\Rightarrow \neg c, \qquad \mathsf{UNKNOWN} \not\Rightarrow I(c).$$

## 1.3 Epistemic separation

Let $B_t$ be verified shared state and $F_t$ speculative state at time $t$. Candidate objects may be generated freely into $F_t$, but promotion is restricted:

$$B_{t+1} = B_t \cup V_t(A_t), \qquad V_t(A_t) := \{a \in A_t : V(a, \Gamma_t, \mathcal{E}_t) = 1\}.$$

This is the mathematical core of BANK/FUTUREBANK separation. Provenance is retained, but provenance does not substitute for verification.

**Proposition 1.1** (Verifier-gated noncontamination)**.** *Suppose the displayed update is the only rule that adds semantic assertions to $B_t$. Then every new assertion in $B_{t+1} \setminus B_t$ has been accepted by $V_t$.*

*Proof.* By the update rule, $B_{t+1} \setminus B_t \subseteq V_t(A_t)$. Membership in $V_t(A_t)$ is defined by verifier acceptance. □

## 1.4 Future-sufficient quotienting

Let $\mathcal{H}$ be a set of search histories and $Q : \mathcal{H} \to \mathcal{S}$ a descriptor. The equivalence relation $h_1 \sim_Q h_2$ iff $Q(h_1) = Q(h_2)$ is computationally useful only if it preserves the future facts relevant to the target. One strong form is

$$Q(h_1) = Q(h_2) \Longrightarrow \mathrm{Future}(h_1) \equiv_V \mathrm{Future}(h_2),$$

where $\equiv_V$ means equality of verifier-relevant continuations, dependency obligations, scope, and certificate-lifting behavior.

**Proposition 1.2** (Sound quotient reuse)**.** *Assume each quotient certificate $\bar{\pi}$ accepted at state $Q(h)$ has a refinement map $L_h$ with*

$$\bar{V}(\bar{\pi}, Q(h)) = 1 \Longrightarrow V(L_h\bar{\pi}, h) = 1.$$

*Then reusing accepted quotient computation cannot introduce a false promoted claim in the original state space.*

*Proof.* Every promotion from a quotient state is refined by $L_h$ and checked by $V$. Soundness is therefore inherited from the stated lifting condition, not from the quotient heuristic itself. □

## 1.5 Guidance and the Ocean gate

A COMPASS-type score may prioritize actions,

$$S(a \mid h) = w^\mathsf{T}\phi(a, h) - \lambda\widehat{\mathrm{Cost}}(a \mid h) - \mu\widehat{\mathrm{Risk}}(a \mid h),$$

but $S$ is not a truth predicate. Candidate search machinery $M$ can itself be tested against a versioned Ocean suite $\mathcal{O}_t$. If $I_1, \ldots, I_k$ are hard invariants, a conservative gate is

$$\mathrm{Acc}(M, \mathcal{O}_t) = 1 \iff \bigwedge_{o \in \mathcal{O}_t} \bigwedge_{j=1}^{k} I_j(M, o) = 1.$$

A verified failure $r_t$ may then be retained by $\mathcal{O}_{t+1} = \mathcal{O}_t \cup \{r_t\}$. Passing a finite gate excludes tested failures; it does not prove universal safety.

# Chapter 2

# DATA-MIND 2.10: Federated Verified Memory

DATA-MIND 2.10 changes where verified information is retained and who sees it by default. It does not change what makes an assertion true. Let $\mathcal{D}$ be the finite set of reasoning departments and let $G_{\mathrm{B}} = (\mathcal{D}, E_{\mathrm{B}})$ be a directed coupling graph. Write $B_t^0$ for the logically shared verified core and $B_t^d$ for department $d$'s local BANK. The implemented read-only view has the abstract form

$$\mathrm{View}_t(d) = B_t^0 \cup B_t^d \cup \bigcup_{(d,e)\in E_{\mathrm{B}}} B_t^e.$$

Thus two departments may share every core item while seeing different local and coupled contexts. The coupling graph is configuration and an experimental variable; it is not itself a mathematical theorem or a security proof.

## 2.1 Four propagation modes

For a verified proposal generated at department $d$, define the physical destination family

$$\Delta_m(d) = \begin{cases} \{d\}, & m = \mathsf{local}, \\ \{d\} \cup N^+_{G_{\mathrm{B}}}(d), & m = \mathsf{coupled}, \\ \{0\}, & m = \mathsf{core}, \\ \{0\} \cup \mathcal{D}, & m = \mathsf{broadcast}. \end{cases}$$

Here destination 0 denotes the shared core. Core is the conservative implementation default: one physical trusted copy becomes logically visible to every department. Broadcast additionally materializes a copy at every local node. Local and coupled modes restrict both storage and ordinary visibility, allowing specialized results to remain near their consumers.

| Mode | Physical storage | Principal research use |
|---|---|---|
| local | source node only | isolate specialized results and measure the value of narrow context |
| coupled | source plus declared neighbors | share interface results along a preregistered topology |
| core | one shared-core copy | retain global reuse without physical duplication |
| broadcast | core plus every local node | reproduce all-copied BANK experiments and stress copying cost |

If $n = |\mathcal{D}|$ and $d^+(d) = |N^+(d)|$, the nominal new-copy counts before deduplication are respectively 1, $1+d^+(d)$, 1, and $1+n$. This is a storage statement, not yet a runtime theorem. A single core copy may still be scanned by many departments, whereas a useful local topology may reduce irrelevant logical visibility.

## 2.2 Truth is invariant under location

Let $\nu_t(x) = 1$ mean that the independent verifier accepts semantic item $x$ in the declared theory and environment. If an accepted axiom or rule is converted to an alternate presentation by a trading callback, assume a certified one-step closure $\mathrm{Tr}(x)$ satisfying

$$\nu_t(x) = 1 \Longrightarrow \forall y \in \mathrm{Tr}(x), \quad \nu_t^\star(y) = 1,$$

where $\nu_t^\star$ checks either the new object directly or a certificate of consequence-closure equivalence. The federated update is then

$$B_{t+1}^j = B_t^j \cup \bigcup_{(x,d,m)\in A_t:\ j\in\Delta_m(d),\ \nu_t(x)=1} (\{x\} \cup \mathrm{Tr}(x)), \qquad j \in \{0\} \cup \mathcal{D}.$$

**Theorem 2.1** (Topology-independent BANK soundness)**.** *Assume every initial BANK item is verifier-accepted or connected to one by a sound certified trade, and the displayed update is the only way semantic items enter any federated BANK node. Then every item visible in every* $\mathrm{View}_t(d)$ *has the same accepted truth status regardless of whether it was propagated locally, to coupled nodes, to the core, or by broadcast.*

*Proof.* Induct on $t$. Existing items satisfy the claim by the induction hypothesis. Each new item is either an accepted proposal $x$ or an element of its certified trade closure. The propagation mode changes only the destination set $\Delta_m(d)$, not $\nu_t$ or $\nu_t^\star$. A view is a union of such BANK nodes, so it contains no item with a new truth status. □

The theorem is deliberately narrow. It proves noncontamination of truth status under the stated transition. It does not prove that every department sees every useful dependency, that the coupling graph is efficient, or that a visible item is authorized for external use.

## 2.3 Dependency-visible reuse

Let $\mathrm{Dep}(x)$ be the immutable dependency set recorded for item $x$. A department can safely reuse a non-self-contained receipt only when

$$x \in \mathrm{View}_t(d) Longrightarrow \mathrm{Dep}(x) \subseteq \mathrm{View}_t(d), \tag{DV}$$

or when the verifier can resolve the dependencies from a separately authenticated content-addressed store. Condition (DV) is stronger than truth of $x$: it is an availability condition for replay.

**Proposition 2.2** (Local replay under dependency visibility). *Suppose every checker invocation for $x$ is deterministic in $(x, \mathrm{Dep}(x), \Gamma, \mathcal{E})$, all dependency identifiers are immutable, and (DV) holds for department $d$. Then a receipt valid when $x$ was deposited remains locally replayable from $\mathrm{View}_t(d)$, provided $\Gamma$ and $\mathcal{E}$ have not changed.*

*Proof.* Condition (DV) supplies every authenticated dependency to the same deterministic checker. Fixing the item, dependencies, theory, and environment fixes the checker input and therefore its result. □

The current 2.10 primitive carries general metadata but does not, by itself, prove (DV) for every application adapter. Dependency closure should therefore be tested as an integration invariant rather than inferred from a successful unit test of propagation.

## 2.4 Three stores and two gates

Federation applies only to trusted BANK. Speculative FUTUREBANK and Sentinel's QUARANTINE remain separate append-only stores. For a candidate $x$, mathematical promotion and operational release are distinct:

$$\mathrm{Trusted}(x) \iff \nu(x) = 1,$$

$$\mathrm{Operational}(x, d) \iff \mathrm{Trusted}(x) \wedge \mathrm{Visible}(x, d) \wedge \mathrm{SecurityApproved}(x, d) \wedge \mathrm{HumanApproved}(x, d)$$

when policy requires human approval. A mathematically valid result may remain quarantined from execution; an unverified result may be preserved in FUTUREBANK or Quarantine without becoming trusted. This orthogonality is inherited from DATA-MIND 2.9 Sentinel and retained by 2.10 [8].

## 2.5 Visibility is not yet confidentiality

The 2.10 view function implements topological visibility, not a full mandatory-access-control lattice. In particular, every core item is logically visible to every registered department. If application security requires labels, let $\lambda(x)$ be an item label and $\kappa(d)$ a department clearance. The application-level view must refine the implementation view:

$$\mathrm{View}_t^{\mathrm{sec}}(d) = \{x \in \mathrm{View}_t(d) : \lambda(x) \preceq \kappa(d) \wedge \mathrm{PurposeAllowed}(x, d)\}.$$

This filtered view is a research obligation for compartmented deployments, not a feature silently attributed to the present code.

# Chapter 3

# Application Map

| Domain | Formal target | Useful certificate | Central preservation obligation |
|---|---|---|---|
| Cryptography | Hardness-relevant invariant or quotient | Proof that a reduction or lifting map preserves the target problem | Do not merge histories with different verifier-relevant futures |
| Vulnerability research | Conformance of implementation to policy | Checked violating trace or inductive proof | Model the complete sufficient routes and system state |
| Critical infrastructure | Bounded compromise propagation | Reachability certificate and non-regression record | Containment must not destroy required availability |
| Secure domains | Authorized boundary crossing and noninterference | Invariant proof or violating trace | Preserve labels, authority, provenance, and declassification rules |
| Software correctness | Safety invariant over reachable states | Inductive proof or finite counterexample | Checker and dependency closure remain fixed |
| Scientific discovery | Predictively sufficient low-dimensional state | Theorem, checked derivation, or preregistered empirical certificate | Preserve the declared observable family, not merely training fit |
| Engineering optimization | Feasible and objective-preserving state merge | Correspondence between feasible continuations | Preserve downstream constraints and optimum values |
| AI safety | Separation of generation, verification, testing, and authority | Gate receipts and invariant certificates | Policy revision must not revise its own truth conditions silently |

| Domain | Natural 2.10 placement pattern | New federation-specific risk |
|---|---|---|
| Cryptography | quotient results local to QH/P/R, reusable certificates in core | cross-scheme reuse without scope metadata |
| Vulnerability research | product-local traces; checker and policy interfaces coupled | target or version contamination |
| Critical infrastructure | subsystem-local models; safety interfaces coupled | coupling topology mistaken for network segmentation |
| Secure domains | compartment-local BANKs plus label-filtered interface nodes | shared core bypasses required clearance |
| Software correctness | module-local lemmas; interface contracts coupled; global invariants in core | visible theorem with unavailable dependencies |
| Scientific discovery | discipline-local hypotheses; replicated findings in core | siloing, duplicated rediscovery, or premature broadcast |
| Engineering optimization | design-team local states; shared constraints coupled or core | local optimum propagated beyond its validity region |
| AI safety | specialized audit nodes around verifier and Sentinel boundaries | truth, visibility, and operational authorization conflated |

The table is not evidence that all eight applications are equally mature. It identifies the object that would need to be certified before speculative target seeking could be treated as defensive knowledge.

# Part II

# Eight Plausible Applications

# Chapter 4

# Cryptography

Discovering unexpected structure in a problem believed to be hard could matter to cryptography. The defensible research question is not whether AMLD can simply try keys faster. It is whether a small, independently checkable invariant or quotient can replace a much larger search without changing the cryptographic problem being solved.

## 4.1 Plausible application

Let $X_\kappa$ be an instance family at security parameter $\kappa$, let $\mathcal{H}_\kappa$ be legal search histories, and let $T(h, a)$ be continuation by action $a$. A quotient candidate $Q_\kappa : \mathcal{H}_\kappa \to \mathcal{S}_\kappa$ induces

$$h_1 \sim_Q h_2 \iff Q_\kappa(h_1) = Q_\kappa(h_2).$$

For defensive significance, equality of quotient states must imply more than similar local features. It should preserve terminal status, admissible continuations, certificate dependencies, and success probability under the declared model. A strong deterministic obligation is a verifier-respecting bisimulation:

$$h_1 \sim_Q h_2 \Longrightarrow \begin{cases} V_{\text{term}}(h_1) = V_{\text{term}}(h_2), \\ \forall a_1 \in A(h_1)\ \exists a_2 \in A(h_2) : T(h_1, a_1) \sim_Q T(h_2, a_2), \\ \forall a_2 \in A(h_2)\ \exists a_1 \in A(h_1) : T(h_1, a_1) \sim_Q T(h_2, a_2). \end{cases}$$

If the search is stochastic, the corresponding condition is equality of the pushed-forward transition kernels on $\mathcal{S}_\kappa$. A COMPASS-like policy may search over candidate descriptors by a score

$$J(Q) = \widehat{\text{reuse}}(Q) - \lambda \log |\operatorname{im} Q| - \mu \widehat{\text{Cost}}(Q) - \nu \widehat{\text{collision risk}}(Q),$$

but promotion requires a certificate for the preservation obligation. Under that certificate, dynamic programming or lemma reuse may occur once per quotient class rather than once per history. The scientific object is therefore a proved structural compression of an admitted problem, not an unverified claim that a named encryption scheme is weak.

**Conjecture 4.1** (Certified quotient advantage)**.** *For some structured cryptographic audit families, there exists a future-sufficient $Q$ for which verified quotient search uses asymptotically fewer distinct* *continuation states than history-level search while returning only liftable certificates.*

*Boundary note* 4.2. No argument in this chapter establishes an efficient attack on AES, RSA, elliptic-curve cryptography, or a post-quantum construction. Hardness is scheme-, model-, and parameter-specific.

## 4.2 DATA-MIND 2.10 update

Federation lets QH retain a verified scheme-specific quotient near the P/R and COMPASS departments that can use it, without forcing that object into every local BANK. Let the immutable scope record be

$$\sigma(x) = (\text{scheme}, \text{version}, \kappa, \Gamma, \mathcal{E}, \text{hash}\,\text{Dep}(x)).$$

A conservative propagation policy is

$$m(x) = \begin{cases} \mathsf{coupled}, & x \text{ is verified but specific to one declared audit family,} \\ \mathsf{core}, & x \text{ carries a checked scope-preserving reduction or lifting theorem,} \\ \mathsf{broadcast}, & \text{only in a preregistered all-copied comparison.} \end{cases}$$

Reuse by department $d$ additionally requires $\sigma(x) \preceq \sigma(\mathfrak{I}_d)$, where the relation means exact agreement on security-relevant scope or a verified translation between scopes. Theorem 2.1 says that copying does not alter truth status; it does not say that a theorem about one scheme or parameter remains applicable to another. Operationally sensitive but mathematically valid findings may pass the verifier while Sentinel still places export or execution behind Quarantine and human authorization. Thus 2.10 can reduce irrelevant sharing and make reuse paths auditable without weakening the boundary between structural cryptographic research and authorization to act on a target.

**Source anchors.** Safe quotienting and COMPASS are developed in *What Checks the Proof?* v6.80, pp. 28–29 and 68 [7]; security as a settlement problem appears in the older security treatise, pp. 42–43 [5].

# Chapter 5

# Cybersecurity and Vulnerability Research

A target-seeking system could generate many suspected traces. Defensively, the important move is to replace the vague target "find a vulnerability" by a model-relative conformance claim and a machine-checkable defect certificate. Speculation remains useful, but it remains quarantined.

## 5.1 Plausible application

Let $\mathcal{M} = (S, S_0, A, \to)$ be a labeled transition system and let $Z$ be its complete finite or infinite traces. A policy defines $\mathrm{Req}(z) \in \{0, 1\}$; the implementation semantics defines $\mathrm{Impl}(z) \in \{0, 1\}$. One-sided acceptance soundness is

$$\Phi_{\text{safe}} := \forall z \in Z\, [\mathrm{Impl}(z) \Rightarrow \mathrm{Req}(z)].$$

If a finite trace $z^\star = (s_0, a_0, \ldots, s_n)$ carries receipts proving

$$s_0 \in S_0, \quad s_i \xrightarrow{a_i} s_{i+1}, \quad \mathrm{Impl}(z^\star) = 1, \quad \mathrm{Req}(z^\star) = 0,$$

then $z^\star$ is a finite $R$-certificate. Conversely, a $P$-certificate may consist of an inductive invariant $I$ satisfying

$$S_0 \subseteq I, \qquad (s \in I \wedge s \to s') \Rightarrow s' \in I, \qquad s \in I \Rightarrow \neg\mathsf{Bad}(s).$$

The $I$-role asks a different question: whether $\Phi_{\text{safe}}$ is independent of the admitted audit theory in a declared metatheory. The $C$-role searches for inconsistent policy obligations. Candidate traces, invariants, and repairs reside in $F_t$; only checker-accepted objects enter $B_t$. For an evolving analyzer $M_t$, Ocean admission can require preservation of four architectural invariants: rejected candidates never enter $B_t$; every derived record has a receipt; dependencies are closed; and resource limits are enforced. This converts vulnerability research into an evidence-producing audit process in which the reasoning system proposes and prioritizes, while a separate checker determines what has actually been established.

**Proposition 5.1** (Counterexample sufficiency)**.** *If the displayed receipts for $z^\star$ verify, then $\Phi_{\text{safe}}$ is* *false in the declared transition model and policy.*

*Proof.* The verified trace instantiates the universal quantifier with an implemented acceptance that is not required, falsifying the implication. □

*Boundary note* 5.2. A checked defect is relative to a model and specification. An incomplete model can miss real behavior; an inadequate policy can be implemented perfectly and still protect the wrong property.

## 5.2 DATA-MIND 2.10 update

The natural federation unit is an audit scope rather than the word "vulnerability." For a checked trace or invariant $x$, record

$$\tau(x) = (\text{product}, \text{build}, \text{configuration}, \text{hash}(\text{Req}), \text{hash}(\text{Impl}), \text{hash}(\mathcal{M})).$$

Product-local P/R nodes can retain scope-specific traces, while a coupled verifier/policy node checks their receipts. Generic checker invariants may enter the core; target-specific evidence normally stays local or coupled. Cross-scope reuse requires either $\tau(x) = \tau(y)$ or a verified simulation $f : \mathcal{M}_y \to \mathcal{M}_x$ satisfying

$$s \to_y s' \Longrightarrow f(s) \to_x f(s'), \qquad \mathsf{Bad}_x(f(s)) \Longrightarrow \mathsf{Bad}_y(s)$$

in the direction needed by the claimed transfer. This prevents a true finding for one build from becoming a falsely scoped finding for another. Federation also sharpens the separation between mathematical evidence and operational capability: a counterexample trace may be verifier-accepted into a restricted BANK node while Sentinel blocks reproduction against a third-party target or quarantines an action requiring risky capabilities. The 2.10 implementation supplies the memory topology; application adapters must still enforce target scope, dependency visibility, authorization, and responsible disclosure.

**Source anchors.** The authentication-defect certificate and four-role audit appear in *What Checks the Proof?* v6.80, pp. 54–55 [7]; the Ocean moat appears in the older treatise, pp. 32–33 [5].

# Chapter 6

# Critical Infrastructure

For hospitals, power systems, water utilities, transportation, and communications, an initial cyber failure can become a physical or availability failure. The natural defensive object is therefore not one secure/insecure bit but a graph of propagation, authority, service, and recovery.

## 6.1 Plausible application

Model the system as a directed labeled graph $G = (V, E, \ell, w)$, where $\ell(e)$ records the type of transition and $w(v) \geq 0$ is a declared consequence weight. For a compromised component $c$, define

$$B_G(c) = \mathrm{Reach}_G(c), \qquad L_G(c) = \sum_{v \in B_G(c)} w(v).$$

A segmentation design $D$ produces $G_D = (V, E_D)$ together with a set $R_D$ of service paths that must remain available. If $E_{D_2} \subseteq E_{D_1}$, reachability monotonicity gives

$$B_{G_{D_2}}(c) \subseteq B_{G_{D_1}}(c), \qquad L_{G_{D_2}}(c) \leq L_{G_{D_1}}(c).$$

But removal of every edge is not a useful defense. Let

$$\mathbf{J}(D) = \left(p_{\mathrm{comp}}(D), \max_c L_{G_D}(c), A_{\mathrm{loss}}(D), C_{\mathrm{verify}}(D), C_{\mathrm{repair}}(D)\right).$$

A candidate dominates another only under a declared partial order and hard service constraints, for example

$$D_2 \succ_X D_1 \iff m_X(D_2) < m_X(D_1) \wedge \forall j \in J_{\mathrm{protected}},\ m_j(D_2) \leq m_j(D_1) \wedge R_{D_2} \models \Psi_{\mathrm{service}}.$$

An AMLD controller can let $P$ prove containment and service obligations, $R$ find checked violating paths, $I$ test whether the obligations are independent of the model, and $C$ detect incompatible requirements. The useful output is a certificate-bearing Pareto comparison, not a single heuristic security score.

**Proposition 6.1** (Edge-removal containment). *If $E' \subseteq E$, then for every $c \in V$, $\mathrm{Reach}_{(V,E')}(c) \subseteq \mathrm{Reach}_{(V,E)}(c)$.*

*Proof.* Every path using only edges in $E'$ is also a path using edges in $E$. □

*Boundary note* 6.2. The proposition proves containment in the graph model only. It does not validate the graph, estimate compromise probability, or show that required services survive segmentation.

## 6.2 DATA-MIND 2.10 update

Federation introduces a second graph that must not be confused with the infrastructure graph. $G_D = (V, E_D)$ models physical, cyber, and service propagation; $G_{\mathrm{B}} = (\mathcal{D}, E_{\mathrm{B}})$ models which reasoning departments see which local BANKs. A narrow bridge in $G_{\mathrm{B}}$ is an information-flow decision, not automatically a firewall or safety barrier in $G_D$.

Subsystem departments may retain detailed local models, couple only verified interface assumptions to neighboring subsystems, and place organization-wide safety requirements in the core. A topology experiment can optimize

$$\mathbf{J}(D, E_{\mathrm{B}}) = \big(p_{\mathrm{comp}}(D), \max_{c} L_{G_D}(c), A_{\mathrm{loss}}(D), C_{\mathrm{scan}}(E_{\mathrm{B}}), C_{\mathrm{copy}}(E_{\mathrm{B}}), R_{\mathrm{miss}}(E_{\mathrm{B}})\big),$$

where $R_{\mathrm{miss}}$ counts requested reuse events blocked by absent interface facts or dependencies. Hard constraints should include service availability, BANK dependency closure, and Sentinel policy. The attractive hypothesis is that local/coupled retention lowers irrelevant scanning while core items preserve universal reuse. The danger is that a poorly chosen coupling hides a safety-critical interface fact. Therefore the coupling graph, topology version, and missed-dependency events must be logged as experimental data, while deployment segmentation remains governed by the separate infrastructure model and operational authorities.

**Source anchors.** Blast radius and narrow bridges are treated in the older security treatise, pp. 10–11; multi-objective security and separation-aware control appear on pp. 53–56 [5].

# Chapter 7

# Military and Intelligence Compartmentation

Compartmentation gives a mathematically clean example of the book's security philosophy: separate domains, restrict crossings, preserve provenance, and never grant speculative reasoning direct operational authority. The formulation below is about defensive authorization and noninterference, not operational attack planning.

## 7.1 Plausible application

Let $(L, \preceq)$ be a security-label lattice and let each state component $x$ carry label $\lambda(x) \in L$. Partition the state space into domains

$$V = \bigsqcup_{i=1}^{k} V_i.$$

For every cross-domain transition $e = (x, y)$, define a boundary predicate $P_e(s, \alpha, \rho)$, where $s$ is current state, $\alpha$ is asserted authority, and $\rho$ is provenance. The safety target is

$$\Phi_{\text{cross}} := \Box\left(\mathsf{Fire}(e, s, \alpha, \rho) \Rightarrow P_e(s, \alpha, \rho)\right).$$

For confidentiality, let $s \equiv_\ell s'$ mean that states agree on observations visible at low label $\ell$. A possibilistic noninterference target is

$$s_0 \equiv_\ell s_0' \Longrightarrow \mathrm{Obs}_\ell(\mathrm{Run}(s_0, u)) = \mathrm{Obs}_\ell(\mathrm{Run}(s_0', u))$$

for every admitted low action sequence $u$, except at explicitly modeled declassification gates. Role $P$ may prove these invariants compositionally; $R$ may supply a checked boundary-crossing or observational counterexample; $I$ and $C$ retain their metatheoretic meanings. Search output does not become trusted merely because a designated role produced it:

$$c \in B_{t+1} \setminus B_t \Longrightarrow \exists \pi\; V(\pi, c, \Gamma, \mathcal{E}) = 1.$$

Thus compartmentation is represented simultaneously as a graph boundary, an authorization predicate, and an observational equivalence, with the verifier mediating any promotion into trusted state.

**Conjecture 7.1** (Cross-domain shared-certificate advantage)**.** *For some compartmented systems, a provenance-preserving shared* BANK *of verified interface lemmas reduces total audit cost relative to isolated audits without weakening domain-local acceptance.*

*Boundary note* 7.2. Noninterference depends on the observation model, declassification rules, scheduler, and environment. The displayed formula is not a universal characterization of national-security systems.

## 7.2 DATA-MIND 2.10 update

This domain most clearly exposes the difference between federation and compartmentation. In the implemented 2.10 topology, every registered department sees the shared core. Therefore a coupling configuration alone is not a clearance mechanism. Define

$$\mathrm{Allowed}(d,x) \iff x \in \mathrm{View}_t(d) \wedge \lambda(x) \preceq \kappa(d) \wedge \mathrm{PurposeAllowed}(d,x).$$

A safe propagation adapter must satisfy

$$j \in \Delta_{m(x)}(d) \Longrightarrow \lambda(x) \preceq \kappa(j), \tag{AC}$$

unless the destination stores only an independently approved redaction or encrypted reference. Because core is visible to all departments, (AC) implies that a labeled item may enter the ordinary core only when every core reader is authorized. Otherwise it remains local, propagates to an authorized coupled subgraph, or is represented by a lower-label interface theorem produced at a declassification gate.

Truth and authority are orthogonal coordinates:

$$(\nu(x), \mathrm{Allowed}(d,x)) \in \{0,1\}^2.$$

A true item can be unauthorized; an authorized channel cannot make an unverified item true. Theorem 2.1 protects the first coordinate, while a label-aware adapter and Sentinel must protect the second. Consequently 2.10 is a useful substrate for compartmented audit research, but the present code should not be described as implementing national-security access control or proving noninterference.

**Source anchors.** Separation, least privilege, and narrow bridges are treated in the older security treatise, pp. 9–11; $P/R/I/C$ security settlement appears on pp. 42–43 [5].

# Chapter 8

# Software Correctness

Software correctness is the most direct beneficial application because both proof and refutation can often be represented by finite machine-checkable objects. Learned guidance may choose what to try; it need not be trusted to decide what counts as a valid proof.

## 8.1 Plausible application

Let $\mathcal{M} = (S, S_0, \to)$ be a transition system and let $I : S \to \{0,1\}$ be a safety invariant. An inductive certificate proves

$$\forall s \in S_0 \; I(s), \qquad \forall s, s' \in S \; [I(s) \wedge s \to s'] \Rightarrow I(s').$$

It follows that $I$ holds throughout $\mathrm{Reach}_{\mathcal{M}}(S_0)$. Role $P$ searches for such an invariant, auxiliary lemmas, and checker receipts. Role $R$ searches for a finite trace

$$s_0 \to s_1 \to \cdots \to s_n, \qquad s_0 \in S_0, \qquad \neg I(s_n).$$

The reasoning architecture is also software and should satisfy its own executable invariants:

$$u \in B_{t+1} \setminus B_t \Rightarrow \exists r \; V(r,u) = 1,$$

$$V(r,u) = 0 \Rightarrow u \notin B_{t+1}, \qquad u \in B_t \Rightarrow \mathrm{Dep}(u) \subseteq B_t.$$

If verifier receipts are content-addressed and dependencies are immutable, replay can check that the accepted object still means what it meant at promotion time. Candidate proofs and candidate program repairs remain in $F_t$ until checking succeeds. An Ocean suite then includes historical counterexamples, malformed certificates, resource-limit tests, and dependency-closure cases. This creates two mutually reinforcing proof obligations: the application program must preserve its safety invariant, and the settlement machinery must preserve the epistemic boundary by which that claim is accepted.

**Proposition 8.1** (Inductive reachability safety)**.** *If the base and step obligations above hold, then every reachable state satisfies $I$.*

*Proof.* Induct on the length of a path from $S_0$. The base case uses $S_0 \subseteq I$; the step uses transition preservation. □

*Boundary note* 8.2. The theorem establishes only the invariant encoded by $I$ in the modeled semantics. It does not prove that the model matches compiled code, hardware, operators, or every environmental condition.

## 8.2 DATA-MIND 2.10 update

For modular software, a department can correspond to a component or proof specialty. Component invariants remain local; verified interface contracts propagate to coupled consumers; global safety theorems enter the core. If component $i$ has assumption $A_i$ and guarantee $G_i$, a federated assume–guarantee obligation is

$$\bigwedge_i (A_i \Rightarrow G_i) \wedge \bigwedge_i \left( \bigwedge_{j \in N_i} G_j \Rightarrow A_i \right) \Longrightarrow \Phi_{\text{system}}.$$

The neighborhood $N_i$ is a semantic dependency set and need not equal the BANK coupling neighborhood. The application adapter should choose $E_{\text{B}}$ so that every contract needed to replay component $i$'s proof is visible, or else resolve immutable dependencies through the core. This is precisely condition (DV).

The four propagation modes create a controlled comparison: local measures isolated modular proof; coupled measures interface-directed reuse; core measures logically global reuse with one copy; and broadcast measures all-copied memory. Correctness is held fixed by the same independent checker in all four conditions. Performance claims require measuring visible BANK size, cache hits, checker replays, missing-dependency faults, proof size, and wall time. A successful federation unit test proves that items flow according to configuration; it does not yet establish that the configured flow is complete for a real compiler, proof assistant, or deployed program.

**Source anchors.** Policy-neutral verification and executable architecture invariants appear in *What Checks the Proof?* v6.80, pp. 10 and 69–70 [7].

# Chapter 9

# Scientific Discovery

The Quotient Hunter idea is scientifically interesting when many apparent histories can be replaced by a smaller state that preserves everything relevant to future prediction or verification. Invariant discovery then becomes a search for sufficient representation, not merely a search for correlation.

## 9.1 Plausible application

Let $H_t$ denote an observed history and $Y_{t:\infty}$ a declared family of future observables. A descriptor $Q(H_t)$ is predictively sufficient when

$$\mathcal{L}(Y_{t:\infty} \mid H_t) = \mathcal{L}(Y_{t:\infty} \mid Q(H_t)).$$

Equivalently, under regularity conditions,

$$Y_{t:\infty} \perp H_t \mid Q(H_t).$$

For deterministic models and observable family $\mathcal{F}$, the obligation may instead be

$$Q(h_1) = Q(h_2) \Rightarrow \forall f \in \mathcal{F}\ \forall \tau \geq 0, \quad f(Y_{h_1}(\tau)) = f(Y_{h_2}(\tau)).$$

Candidate descriptors can be ranked by an information-compression objective

$$U(Q) = \widehat{I}(Q(H_t); Y_{t:\infty}) - \lambda I(Q(H_t); H_t) - \mu \widehat{\mathrm{Cost}}(Q),$$

or by an exact symbolic compression metric when a formal model is available. Yet $U(Q)$ is only a search score. A descriptor enters verified state only after a domain-appropriate certificate: a proof in a formal model, an exact algebraic derivation, or a preregistered empirical test with a clearly limited conclusion. FUTUREBANK may retain failed invariants and conditional “what if” models without asserting them. If a descriptor passes the preservation gate, downstream inference may reuse the same reasoning for every history in a quotient class. The payoff is a certified reduction of effective state dimension and an explicit statement of which observables survive the reduction.

**Conjecture 9.1** (Verifier-history discovery advantage)**.** *On some repeated scientific-model families, retaining rejected quotient hypotheses with provenance reduces expected rediscovery cost without* *increasing false certification, relative to retaining only accepted descriptors.*

*Boundary note* 9.2. Predictive sufficiency is relative to the chosen observables and data-generating assumptions. Good holdout prediction alone is not a theorem that the quotient is universally correct.

## 9.2 DATA-MIND 2.10 update

Scientific departments may represent disciplines, model families, or measurement modalities. A verified algebraic quotient can remain local to QH and the relevant model department; a checked translation theorem can couple it to another discipline; a genuinely scope-independent result may enter the core. Failed or merely suggestive hypotheses remain in FUTUREBANK rather than becoming local "facts." Federation changes the audience of accepted objects, not the evidentiary standard by which an empirical or formal object was accepted.

Let $r_{de} \geq 0$ estimate the preregistered reuse value to department $d$ of items generated by department $e$, and let $c_{de}$ measure visibility or scanning cost. A candidate coupling graph may be scored by

$$U(E_{\mathrm{B}}) = \sum_{(d,e)\in E_{\mathrm{B}}} (r_{de} - \lambda c_{de}) - \mu R_{\mathrm{miss}}(E_{\mathrm{B}}) - \eta R_{\mathrm{leak}}(E_{\mathrm{B}}),$$

subject to dependency visibility and fixed verification rules. This objective can guide a topology experiment, but learned $r_{de}$ values do not certify scientific relevance. Broadcast also carries a special epistemic danger: a model-relative result can appear universal because it is everywhere. Every item should therefore retain observable family, data split, model version, acceptance type, provenance, and dependency hashes. The experimental question is whether selective verified sharing reduces rediscovery cost without raising mis-scoped reuse, not whether federation turns tentative science into theoremhood.

**Source anchors.** FUTUREBANK, planning, and COMPASS appear in *What Checks the Proof?* v6.80, pp. 27–30; future-sufficient quotienting appears on p. 68 [7].

# Chapter 10

# Engineering Optimization

Scheduling, placement, routing, circuit design, materials design, and configuration all contain large combinatorial spaces. Safe state merging can reduce repeated work only when it preserves future constraints and objective values.

## 10.1 Plausible application

Consider

$$\min_{x \in X} f(x) \quad \text{subject to} \quad g_i(x) \le 0 \ (i = 1, \dots, m), \qquad h_j(x) = 0 \ (j = 1, \dots, r).$$

For a partial design history $h$, let

$$\mathcal{C}(h) = \{x \in X : x \text{ extends } h \text{ and satisfies inherited constraints}\}.$$

A strong exact equivalence is

$$h_1 \sim h_2 \iff \exists \varphi : \mathcal{C}(h_1) \to \mathcal{C}(h_2) \text{ bijective such that } f(x) = f(\varphi(x))$$

and all checker-relevant labels are preserved. A weaker quotient may preserve only feasibility and an optimal value:

$$Q(h_1) = Q(h_2) \Rightarrow [\mathcal{C}(h_1) = \varnothing \iff \mathcal{C}(h_2) = \varnothing] \wedge \inf_{x \in \mathcal{C}(h_1)} f(x) = \inf_{x \in \mathcal{C}(h_2)} f(x).$$

Under either certified obligation, dynamic programming may cache downstream value and feasibility by quotient state. A COMPASS-like scheduler may prioritize

$$S(a \mid h) = \alpha \Delta_{\text{goal}} + \beta \rho_Q + \gamma \, \text{reuse} - \delta \widehat{\text{Cost}} - \eta \widehat{\text{Risk}},$$

while feasibility remains checker-defined:

$$\text{Feas}(x) \iff \bigwedge_i g_i(x) \le 0 \wedge \bigwedge_j h_j(x) = 0 \wedge V(x, \Gamma) = 1.$$

Learned guidance therefore decides where to spend computation; it does not redefine admissibility. The research question is whether certified equivalence classes can be discovered cheaply enough that their reuse benefit exceeds the cost of proving them.

**Proposition 10.1** (Optimal-value reuse)**.** *If $Q$ satisfies the displayed optimal-value preservation obligation, then an exact cached value at $Q(h)$ is valid for every history in that quotient class.*

*Proof.* The obligation states equality of the infimum for any two representatives of the same class. □

*Boundary note* 10.2*.* Preserving an optimum value does not necessarily preserve an optimizer, robustness margin, or implementation cost. Those quantities must be added to the equivalence obligation when relevant.

## 10.2 DATA-MIND 2.10 update

Federated BANK supports a modular optimization pattern: design teams retain local feasibility lemmas, couple verified interface constraints to dependent teams, and place system-wide safety envelopes in the core. For item $x$, let $\Omega(x) \subseteq X$ be the certified validity region. Reuse in a partial design state $h$ requires

$$\mathcal{C}(h) \subseteq \Omega(x), \qquad \mathrm{Dep}(x) \subseteq \mathrm{View}_t(d), \qquad \nu(x) = 1.$$

Without the first condition, a true local optimality lemma may be propagated beyond the region in which it is true. Without the second, the receiving department may be unable to replay its receipt.

The BANK topology can itself become a secondary design variable. If $E_{\mathrm{B}}$ determines reuse and copying cost, one may compare

$$\min_{D, E_{\mathrm{B}}} \left(f(D), C_{\mathrm{solve}}(D, E_{\mathrm{B}}), C_{\mathrm{copy}}(E_{\mathrm{B}}), R_{\mathrm{miss}}(E_{\mathrm{B}})\right)$$

subject to the original engineering constraints and topology-independent verification. This does not license changing the coupling graph mid-run until a favorable result appears: topology, adaptation rule, and budget must be frozen or fully logged. Core and broadcast provide reuse-heavy baselines; local and coupled modes test whether selective context improves resource use while preserving the same certified feasible set and objective values.

**Source anchors.** COMPASS and safe quotienting appear in *What Checks the Proof?* v6.80, pp. 28–29 and 68 [7].

# Chapter 11

# AI Safety Itself

A shortcut-generating system is materially different from a shortcut-generating system that is also allowed to declare its own outputs valid. The application here is architectural: separate generation, testing, verification, memory promotion, and operational authority.

## 11.1 Plausible application

Let $G_\theta$ be a mutable generator or search policy and $V_\omega$ a separately governed verifier. For a claim $c$, certificate $\pi$, theory $\Gamma$, and environment $\mathcal{E}$, define

$$\mathsf{Promote}(c, \pi) \iff V_\omega(c, \pi, \Gamma, \mathcal{E}) = 1.$$

Policy-neutral verification requires generator changes not to alter acceptance:

$$\forall \theta_1, \theta_2, c, \pi, \quad V_\omega(c, \pi, \Gamma, \mathcal{E}) \text{ is independent of whether } G_{\theta_1} \text{ or } G_{\theta_2} \text{ proposed } (c, \pi).$$

Candidate self-modifications $\theta'$ remain speculative,

$$\theta' \in F_t \not\Rightarrow \theta' \in B_t,$$

and must pass hard invariants $H_1, \ldots, H_k$ plus an Ocean regression suite:

$$\mathrm{Gate}(\theta') = 1 \iff \left( \bigwedge_{o \in \mathcal{O}_t} \bigwedge_{j=1}^{k} H_j(G_{\theta'}, o) = 1 \right) \wedge V_\omega(\pi_{\theta'}, \theta', \Gamma_{\mathrm{deploy}}) = 1.$$

The gate may also include a statistical endpoint, but statistical success cannot override a failed hard invariant. A verified regression failure $r_t$ updates $\mathcal{O}_{t+1} = \mathcal{O}_t \cup \{r_t\}$. Operational authority is granted by a distinct deployment controller with least privilege and rollback, not by membership in a speculative search frontier. This does not solve alignment in general. It supplies a checkable separation law: policy may change attention, but promotion remains conditional on an independently governed acceptance relation.

**Proposition 11.1** (Generator-independent promotion)**.** *Under the displayed promotion rule, chang-*

*ing $\theta$ alone cannot cause a certificate rejected by $V_\omega$ to enter $B_t$ through the modeled update.*

*Proof.* Promotion depends on $V_\omega$, not on the identity or parameters of the proposing generator. Apply Proposition 1.1. □

*Boundary note* 11.2. Independent verification is a safety mechanism, not a proof of global AI safety. The verifier, specification, interfaces, deployment controller, and environment remain possible failure surfaces.

## 11.2 DATA-MIND 2.10 update

In 2.10, specialized departments can receive different verified contexts around the same trusted core. That supports least-context reasoning experiments: QH may see quotient results, Professor and Child may share evaluation-relevant facts, and COMPASS may combine global verified structure with selected local geometry. Yet the named endpoints are presently topology interfaces, not evidence that all named cognitive modules are fully wired or safe.

For a generated action $a$, a certificate $\pi$, and destination department $d$, define

$$\mathrm{Release}(a,\pi,d)=1 \Longrightarrow \underbrace{V_\omega(a,\pi)=1}_{\text{mathematical or policy certificate}} \wedge \underbrace{a\in \mathrm{View}_t^{\mathrm{sec}}(d)}_{\text{authorized visibility}} \wedge \underbrace{S(a,d)=\mathsf{approve}}_{\text{Sentinel}} \wedge H(a,d),$$

where $H$ is required human authorization for the policy-defined high-risk cases. No propagation mode can delete a conjunct. Core placement does not grant execution authority; local placement does not excuse verification; quarantine does not turn a preserved item into trusted BANK knowledge.

Federation may reduce irrelevant context or isolate resource-pathological strategies, but isolation must not erase the underlying theorem, failure, provenance, or cohort membership. In particular, a future frozen-cohort rerun should preserve the identity of `quartfull` while Sentinel contains an abnormal transaction. This yields a testable safety claim about containment without silent data deletion, not a general alignment theorem.

**Source anchors.** Policy-neutral verification and safe quotienting appear in *What Checks the Proof?* v6.80, pp. 10 and 68 [7]; the Ocean moat appears in the older security treatise, pp. 32–33 [5].

# Part III

# Research Boundaries and Program

# Chapter 12

# A Cross-Domain Experimental Program

The eight sketches can be tested without pretending that they are already established capabilities. DATA-MIND 2.10 creates a new independent variable—BANK topology—while leaving the verifier, Sentinel, FUTUREBANK separation, and settlement outcomes conceptually fixed. For each application family, freeze

$$(\text{target family}, \text{model}, \text{verifier}, \text{budget}, \text{baseline}, \text{split}, \text{stopping rule}, \text{metric}, G_{\mathrm{B}}, m).$$

The first 2.10 comparison should hold the frozen 95% training cohort and learning objective fixed and compare four propagation conditions: core, broadcast, local, and coupled. Local and coupled conditions require a preregistered coupling graph and item-placement rule. The theorem `quartfull`, or any other resource-pathological member, must not be silently removed from the cohort. Sentinel may stop or quarantine an abnormal transaction while the experiment retains its identity, provenance, resource trace, and outcome.

Let $N_{\text{phys}}$ be physical BANK entries, $N_{\text{vis}}$ the mean number visible per active department, and $H_{\text{reuse}}$ the number of useful verified cross-department reuse events. A topology-efficiency coordinate is

$$\eta_{\mathrm{B}} = \frac{H_{\text{reuse}}}{\alpha N_{\text{phys}} + \beta N_{\text{vis}} + \gamma T_{\text{bank}} + \delta R_{\text{miss}}},$$

reported alongside, never instead of, verified settlement. The primary endpoint remains verifier-accepted settlement per unit resource. Secondary endpoints include certificate size, wall time, peak memory, duplicate copy count, visible BANK size, BANK lookup time, reuse latency, missing-dependency events, false-promotion count, Sentinel decisions, quarantine count, and application-specific non-regression coordinates.

## 12.1 Topology comparison

For problem $i$, seed $s$, and mode $m$, record the outcome vector

$$Y_{ism} = (\mathsf{Settle}_{ism}, T_{ism}, M_{ism}, L_{ism}, N^{\mathrm{phys}}_{ism}, N^{\mathrm{vis}}_{ism}, H^{\mathrm{reuse}}_{ism}, R^{\mathrm{miss}}_{ism}, Q_{ism}).$$

Use the same problem–seed blocks across modes. Report paired differences and the full outcome distribution rather than selecting only the best topology after inspection. If adaptive topology is later studied, freeze the adaptation rule and charge its computation to the same budget. A valid result may show that core wins, that selective coupling wins, or that federation overhead dominates; 2.10 makes these alternatives testable rather than predetermining the answer.

## 12.2 A minimum preregistration schema

1. State the target claim and the exact verifier acceptance relation.
2. State which data or instances inform search and which remain held out.
3. State the quotient preservation obligation before observing target results.
4. Freeze the registered departments, directed coupling graph, propagation rule, and any label/clearance filter before the comparison.
5. Match baselines on verifier, budget, and stopping conditions.
6. Record every promoted object with its certificate, dependencies, provenance, and environment.
7. Record physical destination, logical viewers, missing dependencies, Sentinel classification, and Quarantine status for every proposed deposit or release.
8. Treat UNKNOWN as a distinct outcome rather than silently reclassifying it.
9. Retain verified failures as regression cases and report all non-regression coordinates.
10. Preserve every frozen-cohort identity even when a resource tripwire contains its execution.

**Conjecture 12.1** (Separation-aware advantage)**.** *On some repeated target-settlement distributions, a controller that optimizes both settlement progress and verifier-boundary preservation attains a better Pareto frontier than a matched controller that optimizes settlement success alone.*

**Conjecture 12.2** (Selective federation advantage)**.** *For some repeated settlement families with sparse department relevance, a preregistered coupled topology attains the same verifier-accepted settlement set as core access while reducing mean visible BANK mass or total BANK processing cost. For other families with dense reuse, core access may be Pareto-superior. The boundary should be estimated, not assumed.*

# Chapter 13

# Claims This Manuscript Does Not Make

1. No finite Ocean suite proves universal security.
2. No heuristic score is promoted to a truth predicate.
3. No failure to settle is treated as proof of safety, falsity, or independence.
4. No quotient is safe merely because it compresses a benchmark well.
5. No proof of implementation conformance establishes adequacy of the specification.
6. No component-local proof automatically establishes secure composition.
7. No chapter reports a new practical cryptanalytic attack or operational vulnerability.
8. No verifier-gated architecture, by itself, proves global AI safety.
9. No successful federation unit test establishes a training, settlement-rate, wall-time, or memory improvement on theorem benchmarks.
10. No BANK coupling graph, by itself, implements confidentiality, mandatory access control, declassification, or noninterference.
11. No named 2.10 topology endpoint is represented as a completed reasoning module merely because it can receive a federated BANK view.
12. No core placement grants operational authority, and no local placement relaxes verification.

The positive claim is narrower and, for research, more useful: each application can be formulated so that the powerful component proposes search directions while an explicit preservation obligation and independent checker govern promotion. DATA-MIND 2.10 adds a configurable, testable answer to where accepted objects are retained and reused, while leaving truth and security authorization as separate gates. That is the precise point at which this successor extends Security 3 without displacing its older siblings.

# Appendix A

# Glossary

**AMLD**
: A verifier-gated, multi-role architecture for settling formal targets through proof, refutation, independence, contradiction, or an explicit resource-bounded unknown outcome.

**Blast radius**
: The set or weighted consequence of resources reachable after a modeled compromise.

**Certificate**
: A finite object whose validity for a declared claim can be checked independently.

**COMPASS**
: Learned or engineered guidance that ranks search actions without defining truth.

**Coupled BANK**
: A propagation mode that physically retains an accepted item at its source node and the source's explicitly declared neighboring BANK nodes.

**DATA-MIND 2.10**
: The architecture revision that places department-local and coupled verified BANK nodes around one verifier-gated, logically shared core while retaining Sentinel, FUTUREBANK, Quarantine, and verifier sovereignty.

**Federated BANK**
: The family consisting of the verified core, department-local BANK nodes, and their configured read-only views and propagation paths.

**Future-sufficient quotient**
: A state compression that identifies histories only when all declared future facts needed for search and verification are preserved.

**Ocean**
: A versioned adversarial and regression environment used before candidate machinery is granted trusted authority.

**Policy-neutral verification**
: The requirement that changing search policy does not silently change the acceptance relation.

**Quarantine BANK**
Sentinel-controlled preservation for risky, blocked, or security-uncleared objects and actions. Preservation in Quarantine is not promotion into verified BANK.

**Sentinel**
The DATA-MIND 2.9 security and resource governor retained by 2.10 as a gate distinct from mathematical verification.

**Settlement**
Acceptance of a certificate for one of the declared $P/R/I/C$ outcomes.

**Verified BANK**
Trusted state whose semantic assertions enter only through the declared verification gate. In 2.10 it may be retained in the shared core or in local/coupled nodes.

**FUTUREBANK**
Append-only speculative state containing hypotheses, imagined traces, candidate quotients, and proposed repairs not yet admitted to verified BANK. It is distinct from Sentinel Quarantine.

# Appendix B

# Symbols and Notation

| Symbol | Meaning in this manuscript |
|---|---|
| $c$ | Target claim or goal |
| $\Gamma$ | Admitted theory, specification, and scoped dependencies |
| $\mathcal{E}$ | Declared environment or threat model |
| $V$ | Independent verifier or checker |
| $P, R, I, C$ | Prover, Refuter, Independence, and Contradiction settlement roles |
| UNKNOWN | No accepted settlement certificate within declared conditions |
| $B_t$ | Verified BANK contents at time $t$ |
| $B_t^0$ | Logically shared verified core at time $t$ |
| $B_t^d$ | Department $d$'s local verified BANK node at time $t$ |
| $F_t$ | FUTUREBANK speculative contents at time $t$ |
| $\mathcal{D}$ | Set of registered reasoning departments |
| $G_{\mathrm{B}} = (\mathcal{D}, E_{\mathrm{B}})$ | Directed BANK coupling graph |
| $\Delta_m(d)$ | Physical destinations selected by propagation mode $m$ for source $d$ |
| $\mathrm{View}_t(d)$ | Core, local, and coupled verified items visible to department $d$ |
| $\lambda(x), \kappa(d)$ | Application-level security label of item $x$ and clearance of department $d$ |
| $Q$ | Candidate future-sufficient descriptor or quotient map |
| $h_1 \sim_Q h_2$ | Histories have the same quotient state |
| $\mathcal{O}_t$ | Ocean regression suite at time $t$ |
| $\mathrm{Reach}_G(c)$ | Vertices reachable from $c$ in graph $G$ |
| $D_2 \succ_X D_1$ | $D_2$ strictly improves the declared metric for vulnerability class $X$, subject to stated non-regression constraints |

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*AI assistance note. This manuscript develops Brian Tenneson's AMLD, DATA-MIND 2.10 federated BANK, BANK/FUTUREBANK, Sentinel, COMPASS, Quotient Hunter, Ocean, and verifier-gated settlement ideas. ChatGPT assisted with architecture comparison, mathematical formalization, LaTeX drafting, and editorial revision. The named author is responsible for reviewing the definitions, propositions, conjectures, citations, security boundaries, implementation-status statements, and final publication text.*